



Heat Transfer I

Heat Transfer Analysis and Cavity Placement in a Square Chimney

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About this translation

This edition translates the supplied Persian report into English. Source equations, reported data, temperature-array figures, code, and comparison results are retained, with editorial qualifications distinguished from the original numerical statements. The student number is omitted. The original Word drawing overlays are not self-contained raster figures, so the preliminary cavity-location diagrams are reconstructed illustrative grid placements and labeled accordingly.

The MATLAB post-processing mixes Celsius surface temperatures with kelvin ambient temperatures. Consequently, the reported negative heat rates near 18 kW and the resulting optimum are not independently validated physical results. The native files are supplied unchanged; this edition does not silently correct the model or claim a COMSOL rerun.

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1 Geometry Modeling

A grid of 10 cm by 10 cm elements is used. The chimney cavity measures 50 cm by 50 cm, and the outside cross section measures 100 cm by 100 cm. With this element size and the intended minimum wall clearance of 10 cm, sixteen grid-aligned cavity positions can be examined. The source initially describes the clearance as greater than 10 cm; its actual grid includes 10 cm clearances.

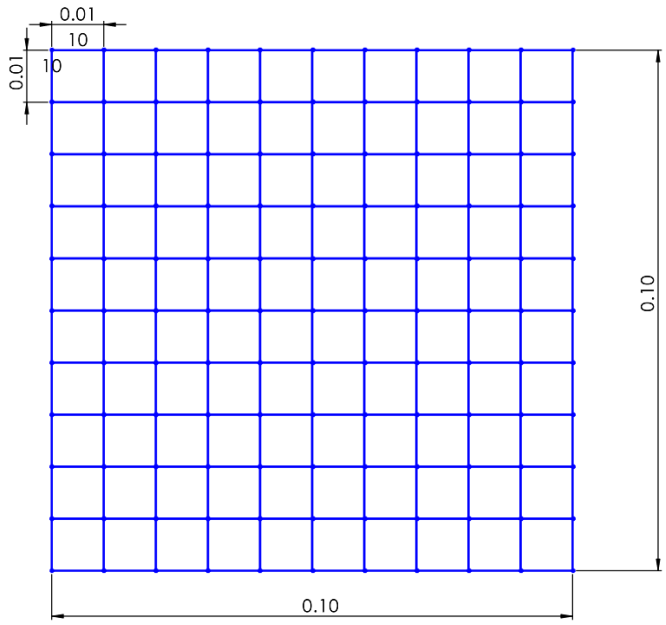


Figure 1: Chimney grid with dimensions in meters.

An initial screening can consider four corner locations and four intermediate locations, followed by refinement in the region with the greatest heat transfer. The diagrams below are reconstructed illustrations of permitted grid placements; the original Word figures combine grid images with drawing overlays.

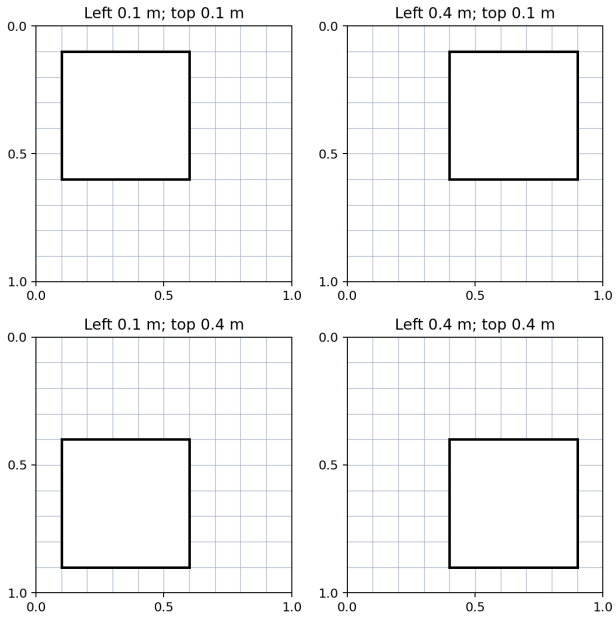


Figure 2: Reconstructed illustrative corner placements on the permitted grid.

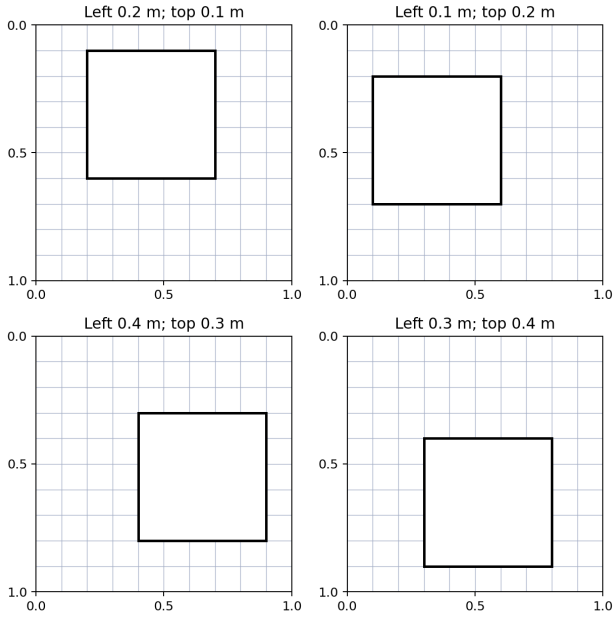


Figure 3: Reconstructed examples of intermediate permitted placements; not a reproduction of Word overlays.

Each of these preliminary locations can be analyzed before evaluating the full sixteen-position grid.

2 Problem Description

The project studies heat transfer through a two-dimensional chimney cross section. The refractory-brick body has a square outside boundary of 1 m by 1 m and contains a square cavity of 0.5 m by 0.5 m. The cavity represents the hot-gas passage, with its wall prescribed at 100 degrees

Celsius. The inner square may be positioned asymmetrically, provided the clearance to every corresponding outer wall is at least 10 cm. Moving the cavity within this region permits study of its influence on temperature distribution.

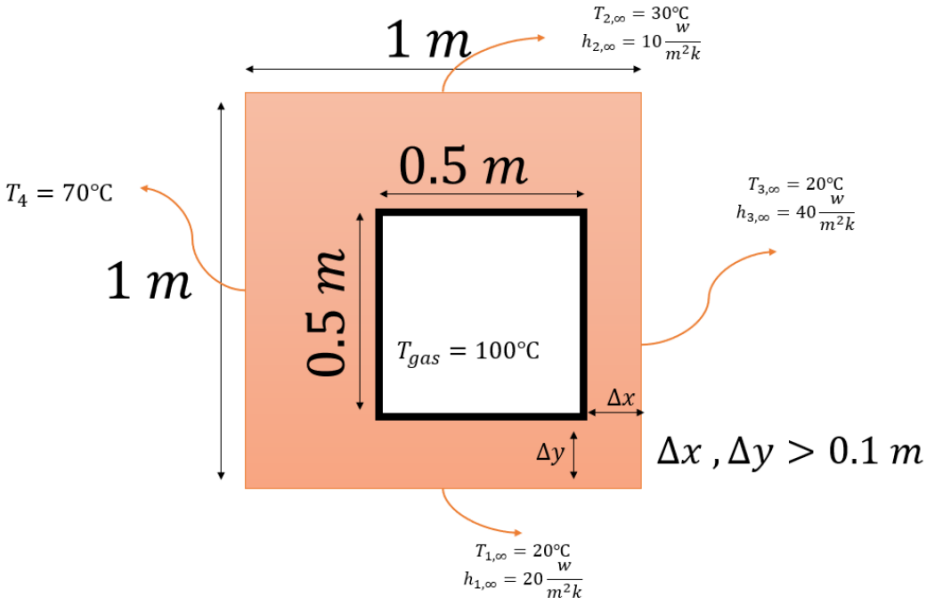


Figure 4: Problem geometry and thermal boundary conditions.

3 Assumptions

The inner cavity wall is maintained at 100°C.

The left outside wall is maintained at 70°C.

The remaining outside walls, top, right, and bottom, exchange heat by convection with their respective surroundings, using coefficients $h_{i,\infty}$ and ambient temperatures $T_{i,\infty}$.

The refractory brick has constant thermal conductivity and is analyzed at steady state.

Material properties are $\rho = 2000$ kg/m^3, $k = 1$ W/(m K), and $C_p = 1000$ J/(kg K). The code uses bottom, top, and right convection coefficients of 20, 10, and 40 W/(m^2 K), respectively, with ambient temperatures of 20, 30, and 20 degrees Celsius.

4 Hand Formulation

The upper-left cavity position is used to illustrate the hand calculation.

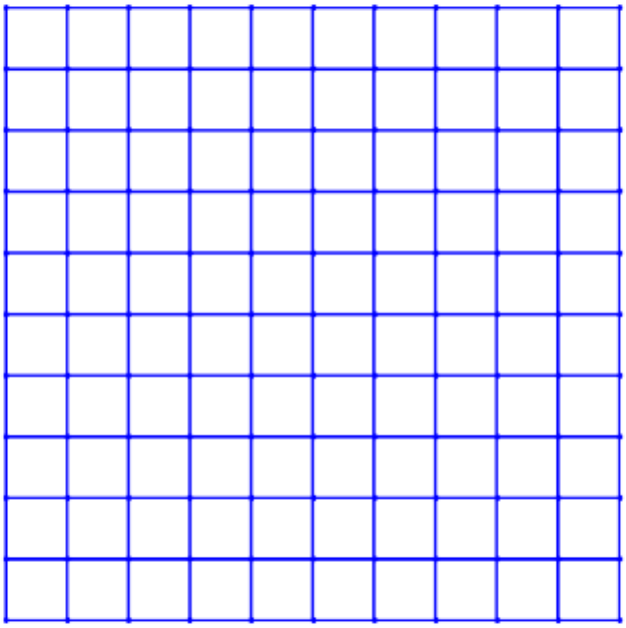


Figure 5: Original grid image for the upper-left location; see reconstructed geometry diagrams.

There are 121 grid nodes. Eleven left-wall nodes have a prescribed temperature of 70 degrees Celsius, and twenty cavity-perimeter nodes have a prescribed temperature of 100 degrees Celsius. Sixteen nodes lie inside the cavity and are not part of the brick. Thus, 74 physical unknown node temperatures remain for each cavity placement.

Nodes are numbered row by row from the upper-left corner: node 1 is at the upper left, node 11 at the upper right, and node 121 at the lower right. For this cavity location, the following nodes share the interior conduction stencil:

19–21, 30–32, 41–43, 52–54, 63–65, 74–76, 79–87, 90–98, and 101–109.

For example, nodes 19 and 109 satisfy

$$T_8 + T_{18} + T_{20} + T_{30} - 4T_{19} = 0.$$

$$T_{98} + T_{108} + T_{110} + T_{120} - 4T_{109} = 0.$$

The general interior-node equation is

$$T_{m-1,n} + T_{m+1,n} + T_{m,n-1} + T_{m,n+1} - 4T_{m,n} = 0.$$

The same equation is written for the other listed interior nodes.

Nodes 2–10 share a boundary balance with one convective face and three conductive connections. The grid spacings are $\Delta x = \Delta y = 0.1$ m, and a unit axial depth $\Delta z = 1$ m is assumed. For node 2,

$$\frac{k\Delta y\Delta z}{2\Delta x}(T_1 - T_2) + \frac{k\Delta y\Delta z}{2\Delta x}(T_3 - T_2) + \frac{k\Delta x\Delta z}{\Delta y}(T_{13} - T_2) + h_{2,\infty}\Delta x\Delta z(T_{2,\infty} - T_2) = 0.$$

The same form applies to bottom nodes 112–120, using $h_{1,\infty}$ and $T_{1,\infty}$ instead. The noncorner right-wall nodes use $h_{3,\infty}$ and $T_{3,\infty}$, with x and y directions exchanged. Because the spacings

are equal, the resulting coefficients have the same pattern. The two right corners are treated separately:

$$\frac{k\Delta y\Delta z}{2\Delta x}(T_{10}-T_{11})+\frac{k\Delta x\Delta z}{2\Delta y}(T_{22}-T_{11}) + \frac{h_{2,\infty}\Delta x\Delta z}{2}(T_{2,\infty}-T_{11})+\frac{h_{3,\infty}\Delta y\Delta z}{2}(T_{3,\infty}-T_{11})=0.$$

$$\frac{k\Delta y\Delta z}{2\Delta x}(T_{120}-T_{121})+\frac{k\Delta x\Delta z}{2\Delta y}(T_{110}-T_{121}) + \frac{h_{1,\infty}\Delta x\Delta z}{2}(T_{1,\infty}-T_{121})+\frac{h_{3,\infty}\Delta y\Delta z}{2}(T_{3,\infty}-T_{121})=0.$$

This gives 74 equations for the 74 physical unknown temperatures. Solving the system in MATLAB gives nodal temperatures for subsequent heat-transfer calculations. Changing the cavity location changes the node indices and boundary assignments rather than introducing a different type of governing equation.

5 MATLAB Solution

The linear system can be solved efficiently, allowing all sixteen grid-aligned cavity positions to be examined. The coefficient, unknown, and known vectors are arranged as

$$A\mathbf{T}=\mathbf{C}.$$

The code retains a full 121-by-121 matrix A and 121-by-1 vectors T and C, prescribing known grid values to close the system. The formal matrix-inverse expression in the source is

$$\mathbf{T}=A^{-1}\mathbf{C}.$$

5.1 Original MATLAB code

```
clc
clear

k=1; % W/m.k

h1=20; % W/m^2.k
h2=10;
h3=40;

T1=20+273; % Kelvin
T2=30+273;
T3=20+273;
Tg=100+273; % Gas temperature
T4=70+273; % Left wall temperature

x=0.1; % meter
y=0.1;
z=1;

allst=cell(1,16); % For all states of the hole
Qall=zeros(16,1); % Total heat flux for all states of the hole
for g=2:5
    for h=2:5
        TT=zeros(11);
        %Left wall
```

```
        TT(:,1)=T4;
        %Inner chamber
        TT(g:g+5,h:h+5)=Tg;

        T=zeros(121,1);
        A=zeros(121);
        C=zeros(121,1);
        for n=1:11
            for m=1:11
                i=(n-1)*11+m;
                f=0; % flag variable
                if TT(n,m)==0
                    % Upper wall nodes 2-10
                    if n==1 && m<11
                        A(i,i)=-(2*k+x*h2);
                        A(i,i-1)=k/2;
                        A(i,i+1)=k/2;
                        A(i,i+11)=k;
                        C(i,1)=-h2*x*T2;
                        f=1;
                    end

                    % Lower wall nodes 112-120
                    if n==11 && m<11
                        A(i,i)=-(2*k+x*h1);
                        A(i,i-1)=k/2;
                        A(i,i+1)=k/2;
                        A(i,i-11)=k;
                        C(i,1)=-h1*x*T1;
                        f=1;
                    end

                    % Right wall nodes 22,33,44,55,66,77,88,99,110
                    if m==11 && n<11 && n>1
                        A(i,i)=-(2*k+y*h3);
                        A(i,i-11)=k/2;
                        A(i,i+11)=k/2;
                        A(i,i-1)=k;
                        C(i,1)=-h3*y*T3;
                        f=1;
                    end

                    % Upper-right corner node 11
                    if n==1 && m==11
                        A(i,i)=-(k+(x/2)*(h2+h3));
                        A(i,i-1)=k/2;
                        A(i,i+11)=k/2;
                        C(i,1)=-(h2*(x/2)*T2 + h3*(y/2)*T3);
                        f=1;
                    end

                    % Lower-right corner node 121
                    if n==11 && m==11
                        A(i,i)=-(k+(x/2)*(h1+h3));
                        A(i,i-1)=k/2;
                        A(i,i-11)=k/2;
                        C(i,1)=-(h1*(x/2)*T1 + h3*(y/2)*T3);
                        f=1;
                    end

                    % All the nodes in the brick (only conduction)
                    if f==0
                        A(i,i)=-4;
                        A(i,i-1)=1;
                        A(i,i+1)=1;
                        A(i,i-11)=1;
                        A(i,i+11)=1;
                    end

                else
                    T(i)=TT(n,m);
                    A(i,i)=1;
                    C(i,1)=TT(n,m);
```

```
                end
            end
        end

        TK=A\C; %Solve for kelvin
        TC=zeros(121,1); % Celsius
        for j=1:121
            TC(j,1)=TK(j,1)-273;
        end

        Vis=zeros(11,11);
        for i=1:11
            for j=1:11
                Vis(i,j)=TC((i-1)*11+j,1);
            end
        end
        allst(4*(g-2)+h-1)=mat2cell(Vis,11,11);


        % Calculate the total heat transfer of the outer boundary

        % Four corners
        q1= h2*(x/2)*z*(T4-T2); % Node 1 (corner)
        q111= h1*(x/2)*z*(T4-T1); % Node 111 (corner)
        q11= h2*(x/2)*z*(Vis(1,11)-T2) + h3*(y/2)*z*(Vis(1,11)-T3); % Node 11(corner)
        q121= h1*(x/2)*z*(Vis(11,11)-T1) + h3*(y/2)*z*(Vis(11,11)-T3); % Node 121(corner)
        q=q1+q11+q111+q121;

        % 3 Sides (Up, Right, Down)
        for i=2:10
            qup=h2*x*z*(Vis(1,i)-T2);
            qdown=h1*x*z*(Vis(11,i)-T1);
            qright=h3*y*z*(Vis(i,11)-T3);
            q=q+qup+qdown+qright;
        end
        Qall(4*(g-2)+h-1,1)=q/1000; % kW
    end
end
maxq=min(Qall)
```

A cell array stores the sixteen 11-by-11 temperature arrays.

	1	2	3	4	5	6	7	8	9	10	11
1	70	65.8575	65.1450	65.0125	64.9299	64.5672	62.4732	50.2719	40.7541	32.9448	23.7631
2	70	100	100	100	100	100	100.0000	69.2020	50.6539	36.5758	23.3966
3	70	100	100	100	100	100	100	75.8822	56.0836	39.3080	23.8446
4	70	100	100	100	100	100	100	78.2433	58.4904	40.7279	24.1230
5	70	100	100	100	100	100	100	78.6006	58.9067	40.9903	24.1758
6	70	100	100	100	100	100	100	77.2525	57.5454	40.1509	24.0054
7	70	100	100	100	100	100	100	72.8640	53.8713	38.0627	23.5872
8	70	80.0337	82.0187	81.8383	80.8000	78.6263	73.6179	60.3324	47.0130	34.6414	22.9151
9	70	68.1160	66.2026	64.5345	62.7355	60.0874	55.5130	47.8346	39.2070	30.5748	22.1110
10	70	56.2277	50.1414	47.3617	45.5202	43.4747	40.5120	36.2860	31.4056	26.3399	21.2678
11	70	36.6536	30.7734	29.2507	28.5088	27.7791	26.7744	25.3920	23.7893	22.1114	20.4224

Figure 6: Original MATLAB temperature array for State 1.

	1	2	3	4	5	6	7	8	9	10	11
1	70	58.2498	63.8414	64.7988	64.9516	64.9108	64.5131	62.1680	48.4949	36.7594	24.5077
2	70	77.8288	100	100	100	100	100	100.0000	66.0211	43.7768	24.7943
3	70	83.0653	100	100	100	100	100	100	71.8128	47.5324	25.4704
4	70	84.4326	100	100	100	100	100	100	73.6978	49.0694	25.7863
5	70	84.6650	100	100	100	100	100	100	73.9088	49.2611	25.8267
6	70	84.2273	100	100	100	100	100	100	72.6764	48.2395	25.6114
7	70	82.2443	100	100	100	100	100	100	68.5574	45.4091	25.0316
8	70	74.7498	80.3570	81.4827	81.2461	80.1791	77.7273	71.8534	56.1440	39.8079	23.9499
9	70	66.3981	65.1953	64.3277	63.3226	61.7429	58.8769	53.5424	44.3572	33.7286	22.7511
10	70	55.6471	49.6985	47.3103	45.9736	44.5930	42.4950	39.0819	34.0139	27.9981	21.6065
11	70	36.4919	30.6413	29.2413	28.6686	28.1604	27.4283	26.2763	24.6184	22.6432	20.5312

Figure 7: Original MATLAB temperature array for State 2.

	1	2	3	4	5	6	7	8	9	10	11
1	70	53.9947	55.3885	63.3504	64.7140	64.9333	64.8861	64.3834	61.4145	44.1034	25.9635
2	70	69.2897	77.4930	100	100	100	100	100	100.0000	58.6211	27.6412
3	70	75.6712	85.2936	100	100	100	100	100	100	62.7398	28.4889
4	70	78.1015	88.0104	100	100	100	100	100	100	63.8493	28.7456
5	70	78.7243	88.6464	100	100	100	100	100	100	63.9118	28.7601
6	70	78.1494	87.8510	100	100	100	100	100	100	63.0376	28.5524
7	70	76.0222	84.6082	100	100	100	100	100	100	59.6863	27.7940
8	70	71.3310	74.5597	79.5742	80.7765	80.5884	79.3224	76.1875	68.5156	47.9136	25.6025
9	70	64.7423	62.7255	62.9606	62.9433	62.2549	60.5138	56.9120	49.9613	37.8500	23.6092
10	70	54.9125	48.6394	46.5994	45.7814	44.9739	43.5658	40.9854	36.5678	29.9157	22.0080
11	70	36.2682	30.3204	29.0160	28.6090	28.2934	27.7901	26.8962	25.4086	23.2369	20.6556

Figure 8: Original MATLAB temperature array for State 3.

	1	2	3	4	5	6	7	8	9	10	11
1	70	51.6254	49.9427	54.3951	63.1796	64.6825	64.9154	64.8099	63.9441	58.8545	29.1832
2	70	64.9049	66.8180	76.6241	100	100	100	100	100	100.0000	35.4279
3	70	71.1762	75.8001	85.2835	100	100	100	100	100	100	35.9520
4	70	74.0000	79.9225	88.7099	100	100	100	100	100	100	35.9956
5	70	74.9011	81.1802	89.6334	100	100	100	100	100	100	35.9952
6	70	74.4241	80.2639	88.6435	100	100	100	100	100	100	35.9472
7	70	72.5315	76.8079	84.6767	100	100	100	100	100	100	35.3712
8	70	68.8938	69.7596	73.2554	78.6355	80.0006	79.6610	77.7485	72.9461	61.0160	28.5068
9	70	63.2840	60.0812	59.9497	61.2860	61.7058	60.8947	58.3871	53.0198	42.6112	24.6782
10	70	54.1611	47.3316	45.1763	44.8530	44.6419	43.8251	41.8853	38.1349	31.7306	22.4093
11	70	36.0288	29.9079	28.5710	28.3076	28.1837	27.8785	27.1941	25.9039	23.7672	20.7721

Figure 9: Original MATLAB temperature array for State 4.

	1	2	3	4	5	6	7	8	9	10	11
1	70	55.7865	53.6298	53.2096	52.8453	51.9336	49.5150	43.9866	37.9972	31.8350	23.5549
2	70	75.5445	76.3914	76.3913	75.9643	74.6207	70.5848	58.2036	46.0810	34.7288	23.0493
3	70	100	100	100	100	100	100.0000	72.1620	53.3943	37.9501	23.5789
4	70	100	100	100	100	100	100	77.0500	57.3840	40.0983	23.9972
5	70	100	100	100	100	100	100	78.6541	58.9935	41.0619	24.1909
6	70	100	100	100	100	100	100	78.5729	58.8742	40.9646	24.1703
7	70	100	100	100	100	100	100	76.7635	56.9656	39.7522	23.9230
8	70	100	100	100	100	100	100	71.5154	52.4724	37.1558	23.4014
9	70	76.9774	77.6031	77.1101	76.2243	74.4075	69.8293	56.8257	44.2530	32.9969	22.5823
10	70	60.3065	56.3248	54.6132	53.3795	51.5765	48.0838	41.7050	34.7170	27.9966	21.5925
11	70	37.9237	32.7765	31.6384	31.1041	30.4352	29.2245	27.1935	24.9135	22.6801	20.5341

Figure 10: Original MATLAB temperature array for State 5.

	1	2	3	4	5	6	7	8	9	10	11
1	70	53.7070	52.6323	52.9096	52.9913	52.6979	51.6525	48.7776	42.1047	34.2119	24.0323
2	70	69.8048	74.5885	75.9171	76.1701	75.7719	74.2196	69.4542	54.8195	39.5673	24.0143
3	70	80.9238	100	100	100	100	100	100	68.1517	45.2234	25.0049
4	70	83.8904	100	100	100	100	100	100	72.5640	48.1696	25.5982
5	70	84.6378	100	100	100	100	100	100	73.9345	49.2930	25.8338
6	70	84.6608	100	100	100	100	100	100	73.8810	49.2342	25.8208
7	70	84.0054	100	100	100	100	100	100	72.3554	47.9420	25.5477
8	70	81.3610	100	100	100	100	100	100	67.5986	44.6307	24.8678
9	70	71.4384	75.8510	76.6792	76.5554	75.8055	73.8220	68.5370	53.4084	38.1143	23.6049
10	70	58.5415	55.2867	54.3102	53.7370	52.8447	50.9456	46.9174	39.3836	30.8133	22.1626
11	70	37.4409	32.4440	31.5378	31.2378	30.8907	30.1982	28.8036	26.3954	23.5927	20.7194

Figure 11: Original MATLAB temperature array for State 6.

	1	2	3	4	5	6	7	8	9	10	11
1	70	51.9817	49.9468	51.7048	52.6271	52.8419	52.4704	51.0650	47.1523	37.8256	24.7796
2	70	65.9716	67.9971	73.8274	75.6078	75.9768	75.4577	73.3836	67.0117	47.5108	25.6315
3	70	73.9077	82.2427	100	100	100	100	100	100.0000	59.5743	27.7772
4	70	77.4164	87.0658	100	100	100	100	100	100	63.0093	28.5464
5	70	78.6923	88.6041	100	100	100	100	100	100	63.9163	28.7613
6	70	78.7486	88.6585	100	100	100	100	100	100	63.8948	28.7560
7	70	77.6436	87.2812	100	100	100	100	100	100	62.9068	28.5211
8	70	74.5446	82.8226	100	100	100	100	100	100	59.2113	27.6839
9	70	67.7123	69.4647	74.5967	75.9990	76.1082	75.2737	72.7534	65.8991	46.2546	25.2628
10	70	56.8399	52.7272	52.9229	53.2913	53.1601	52.2331	49.8408	44.5882	34.6453	22.9601
11	70	36.9202	31.6814	31.0763	31.0833	31.0077	30.6578	29.7884	27.9679	24.7782	20.9673

Figure 12: Original MATLAB temperature array for State 7.

	1	2	3	4	5	6	7	8	9	10	11
1	70	50.7043	47.4756	48.6849	51.2978	52.4392	52.6172	51.9242	49.5230	42.7832	25.9042
2	70	63.3752	62.7321	66.6679	73.3315	75.3602	75.6699	74.7024	71.2154	60.6361	28.5459
3	70	70.0644	73.4097	81.9232	100	100	100	100	100	100.0000	35.3744
4	70	73.4729	78.9191	87.6150	100	100	100	100	100	100	35.9475
5	70	74.9079	81.1788	89.6179	100	100	100	100	100	100	35.9952
6	70	74.9800	81.2702	89.6779	100	100	100	100	100	100	35.9950
7	70	73.7417	79.2441	87.8233	100	100	100	100	100	100	35.9449
8	70	70.7429	74.1412	82.3713	100	100	100	100	100	100	35.3436
9	70	65.0887	64.2065	67.5208	73.6576	75.4494	75.5374	74.2109	70.2400	59.4047	28.1783
10	70	55.4052	50.0755	49.8476	51.6602	52.6025	52.4893	51.0660	47.3446	39.2004	23.9863
11	70	36.4566	30.8427	30.1339	30.5332	30.8112	30.7511	30.2194	28.8718	26.0662	21.2566

Figure 13: Original MATLAB temperature array for State 8.

	1	2	3	4	5	6	7	8	9	10	11
1	70	51.9517	48.2994	47.2910	46.5773	45.4372	43.3389	39.8902	35.6877	30.7445	23.3436
2	70	66.7055	65.2769	64.4345	63.3679	61.3534	57.3530	50.1574	41.7456	32.7177	22.6608
3	70	79.5934	81.6681	81.8023	81.1064	79.2556	74.5623	61.6408	48.4196	35.7201	23.1500
4	70	100	100	100	100	100	100.0000	73.4239	54.5720	38.5932	23.6992
5	70	100	100	100	100	100	100	77.4826	57.8513	40.3813	24.0537
6	70	100	100	100	100	100	100	78.6554	58.9692	41.0272	24.1831
7	70	100	100	100	100	100	100	78.1700	58.3427	40.5751	24.0890
8	70	100	100	100	100	100	100	75.6818	55.6565	38.8416	23.7345
9	70	100	100	100	100	100	100	68.9006	49.7600	35.4004	23.0420
10	70	69.8762	68.6875	68.0877	67.5661	66.3613	62.5847	50.1605	39.0827	29.9579	21.9689
11	70	40.8173	36.7862	36.0971	35.8155	35.2944	33.8172	30.0738	26.4523	23.3796	20.6686

Figure 14: Original MATLAB temperature array for State 9.

	1	2	3	4	5	6	7	8	9	10	11
1	70	51.2809	47.8108	47.1549	46.8317	46.2180	44.9024	42.3290	37.9583	32.1651	23.6330
2	70	64.9374	64.2144	64.1434	63.8087	62.7869	60.4338	55.5565	46.6278	35.6997	23.2656
3	70	74.2542	79.9661	81.3956	81.4729	80.6872	78.4892	72.8356	57.2968	40.7401	24.1553
4	70	82.1134	100	100	100	100	100	100.0000	68.9837	45.8087	25.1181
5	70	84.1995	100	100	100	100	100	100	72.8291	48.3929	25.6447
6	70	84.6844	100	100	100	100	100	100	73.9399	49.2890	25.8326
7	70	84.5381	100	100	100	100	100	100	73.6414	48.9907	25.7680
8	70	83.4680	100	100	100	100	100	100	71.6349	47.2645	25.4023
9	70	79.3340	100	100	100	100	100	100	65.6338	43.0300	24.5309
10	70	63.8679	66.9059	67.6364	67.7296	67.3943	66.1146	61.8781	47.8701	34.6908	22.9080
11	70	39.2319	36.1192	35.9100	35.8877	35.7328	35.1862	33.5275	29.2779	24.9553	20.9829

Figure 15: Original MATLAB temperature array for State 10.

	1	2	3	4	5	6	7	8	9	10	11
1	70	50.5467	46.7447	46.4811	46.6177	46.4510	45.6960	43.9145	40.3308	33.9590	24.0085
2	70	63.2676	61.7202	62.7622	63.3871	63.1962	61.9051	58.7302	52.0556	39.7074	24.1004
3	70	70.8036	74.1065	79.4602	80.9722	81.0416	79.9981	77.0456	69.4540	48.7147	25.7815
4	70	75.8404	84.4418	100	100	100	100	100	100.0000	59.9160	27.8484
5	70	78.1161	87.8204	100	100	100	100	100	100	63.1008	28.5677
6	70	78.8037	88.7236	100	100	100	100	100	100	63.9196	28.7620
7	70	78.3750	88.2704	100	100	100	100	100	100	63.8157	28.7369
8	70	76.4261	85.9830	100	100	100	100	100	100	62.6064	28.4491
9	70	71.3462	79.2355	100	100	100	100	100	100	58.1606	27.4401
10	70	59.7234	59.6128	65.4629	67.1630	67.5395	67.1981	65.6081	60.2681	42.5959	24.5105
11	70	37.9345	34.0295	35.0756	35.6498	35.7967	35.6449	34.9662	32.8683	27.4443	21.4944

Figure 16: Original MATLAB temperature array for State 11.

	1	2	3	4	5	6	7	8	9	10	11
1	70	49.8748	45.5515	45.1647	45.8185	46.1738	45.8886	44.6529	41.7662	35.8499	24.4310
2	70	61.8487	59.1346	59.8091	61.7862	62.6680	62.2523	60.1314	55.0472	44.4510	25.1669
3	70	68.3853	69.3293	73.1508	78.8492	80.4596	80.3213	78.5732	73.8401	61.7401	28.6693
4	70	72.3632	76.6464	84.6158	100	100	100	100	100	100.0000	35.3848
5	70	74.4210	80.2773	88.6661	100	100	100	100	100	100	35.9483
6	70	75.0435	81.3758	89.7712	100	100	100	100	100	100	35.9952
7	70	74.3772	80.4112	89.0431	100	100	100	100	100	100	35.9946
8	70	72.0539	76.8488	85.9898	100	100	100	100	100	100	35.9394
9	70	66.9897	68.9403	78.0673	100	100	100	100	100	100	35.2788
10	70	56.9646	53.8554	57.3392	64.7380	66.9005	67.3367	66.7408	64.1842	55.6457	27.4058
11	70	37.0133	32.1775	32.6960	34.7124	35.5274	35.7054	35.4422	34.3504	30.9927	22.2998

Figure 17: Original MATLAB temperature array for State 12.

	1	2	3	4	5	6	7	8	9	10	11
1	70	50.1256	45.3926	43.8113	42.7865	41.5500	39.7337	37.1654	33.9360	29.8387	23.1650
2	70	62.6806	59.2094	57.3444	55.6787	53.3900	49.8435	44.6613	38.3061	30.9656	22.3161
3	70	71.3872	71.4200	70.6781	69.1940	66.4878	61.5889	53.3302	43.6614	33.4013	22.6975
4	70	81.4482	84.4055	84.7538	83.9316	81.7784	76.6941	63.4093	49.6079	36.2810	23.2509
5	70	100	100	100	100	100	100	74.0048	55.0802	38.8636	23.7519
6	70	100	100	100	100	100	100	77.5296	57.8444	40.3414	24.0442
7	70	100	100	100	100	100	100	78.2693	58.4263	40.6133	24.0962
8	70	100	100	100	100	100	100	77.1213	56.9784	39.5891	23.8837
9	70	100	100	100	100	100	100	73.2375	52.7769	36.8811	23.3301
10	70	100	100	100	100	100	100	63.0517	44.0106	31.8283	22.3158
11	70	49.6303	47.0427	46.7114	46.6487	46.4785	45.1796	34.9586	28.3857	24.1056	20.8027

Figure 18: Original MATLAB temperature array for State 13.

	1	2	3	4	5	6	7	8	9	10	11
1	70	49.9002	45.2120	43.8280	43.0837	42.2075	40.8122	38.5755	35.2550	30.6889	23.3391
2	70	62.0946	58.7718	57.3361	56.2333	54.6745	52.0450	47.6930	41.1327	32.7697	22.6850
3	70	69.7065	70.4446	70.5112	69.8390	68.2121	65.0005	59.0188	48.8133	36.5720	23.3416
4	70	76.2866	82.7890	84.4249	84.3996	83.3344	80.7258	74.5686	58.5296	41.3635	24.2702
5	70	82.6508	100	100	100	100	100	100	69.3729	46.0821	25.1736
6	70	84.3166	100	100	100	100	100	100	72.8800	48.4183	25.6494
7	70	84.6157	100	100	100	100	100	100	73.7289	49.0617	25.7829
8	70	84.1463	100	100	100	100	100	100	72.9738	48.3169	25.6221
9	70	81.9695	100	100	100	100	100	100	69.8494	45.6098	25.0480
10	70	73.7317	100	100	100	100	100	100	60.8140	39.2251	23.7340
11	70	42.9574	46.1955	46.6064	46.6558	46.6401	46.4651	45.0808	34.1814	26.7427	21.3096

Figure 19: Original MATLAB temperature array for State 14.

	1	2	3	4	5	6	7	8	9	10	11
1	70	49.5743	44.7470	43.5023	43.0189	42.4579	41.4316	39.5871	36.4952	31.6138	23.5328
2	70	61.3493	57.7026	56.6240	56.0765	55.1485	53.2722	49.7978	43.8851	34.8275	23.1157
3	70	68.1204	68.0902	69.2146	69.5146	68.7875	66.7108	62.4470	54.4201	40.6952	24.2002
4	70	73.0422	77.3231	82.6294	83.9800	83.7760	82.3365	78.8591	70.6531	49.3331	25.8961
5	70	76.7252	85.5307	100	100	100	100	100	100.0000	60.0879	27.8872
6	70	78.3280	88.0745	100	100	100	100	100	100	63.1314	28.5749
7	70	78.5121	88.4393	100	100	100	100	100	100	63.8629	28.7482
8	70	77.2813	87.1705	100	100	100	100	100	100	63.5718	28.6779
9	70	73.4427	82.9614	100	100	100	100	100	100	61.7465	28.2431
10	70	63.5281	71.2325	100	100	100	100	100	100	55.1712	26.7461
11	70	39.4371	38.4405	45.6217	46.5335	46.6460	46.6349	46.4329	44.8281	32.1922	22.3673

Figure 20: Original MATLAB temperature array for State 15.

	1	2	3	4	5	6	7	8	9	10	11
1	70	49.2129	44.1301	42.8424	42.5688	42.3056	41.5983	40.0860	37.2833	32.4179	23.7085
2	70	60.5736	56.3628	55.1777	55.1325	54.8332	53.5991	50.8171	45.5980	36.7578	23.5414
3	70	66.7188	65.5696	66.3733	67.9502	68.2956	67.1477	63.9855	57.5337	45.4740	25.2722
4	70	70.7321	72.8236	76.7956	81.9993	83.2514	82.7106	80.4433	75.0773	62.3322	28.7775
5	70	73.3860	78.1969	85.9862	100	100	100	100	100	100	35.3939
6	70	74.6151	80.5919	88.9524	100	100	100	100	100	100	35.9491
7	70	74.4826	80.6031	89.2316	100	100	100	100	100	100	35.9952
8	70	72.7121	78.1062	87.3708	100	100	100	100	100	100	35.9931
9	70	68.2595	71.7389	82.1456	100	100	100	100	100	100	35.9216
10	70	58.5870	58.4444	69.4725	100	100	100	100	100	100	35.0660
11	70	37.6441	33.9791	37.3001	45.4769	46.5148	46.6418	46.6196	46.3147	43.8982	24.8705

Figure 21: Original MATLAB temperature array for State 16.

The source identifies State 5 as the location with the greatest outward heat-transfer magnitude and reports approximately 18.167 kW. The original calculation uses a signed negative indicator and selects its minimum; the temperature-unit issue is discussed separately in the editorial appendix.

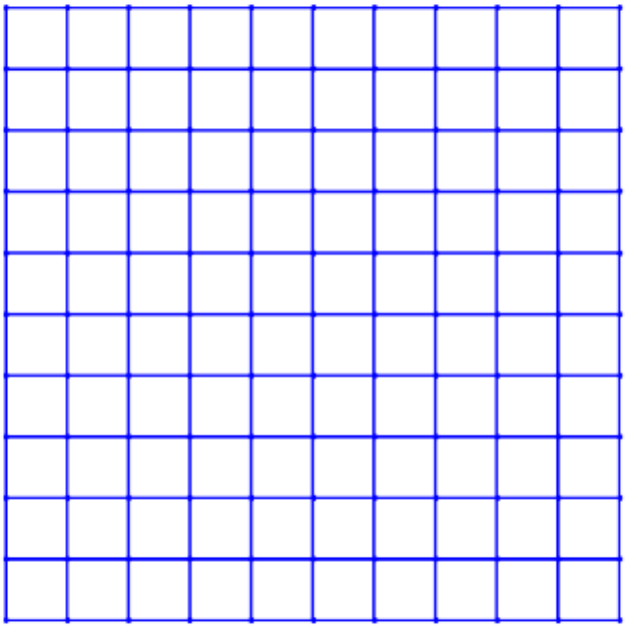


Figure 22: Original State 5 grid image; its cavity drawing overlay is not embedded in the raster.

The source places the largest reported heat-transfer magnitude near State 5. It notes that the 10 cm grid is coarse and cannot establish a continuous global optimum. Refining the grid would better resolve the location but increase computation. The source identifies the region near the left wall, between States 1, 5, and 9, as promising, then compares the temperature results with COMSOL.

	1
1	-18.1231
2	-18.0523
3	-17.9550
4	-17.7513
5	-18.1666
6	-18.0967
7	-17.9975
8	-17.7889
9	-18.1465
10	-18.0747
11	-17.9746
12	-17.7664
13	-18.0338
14	-17.9554
15	-17.8552
16	-17.6528

Figure 23: Original signed heat-rate indicators for States 1–16.

6 COMSOL Model and Comparison

State 1 is modeled first in COMSOL.

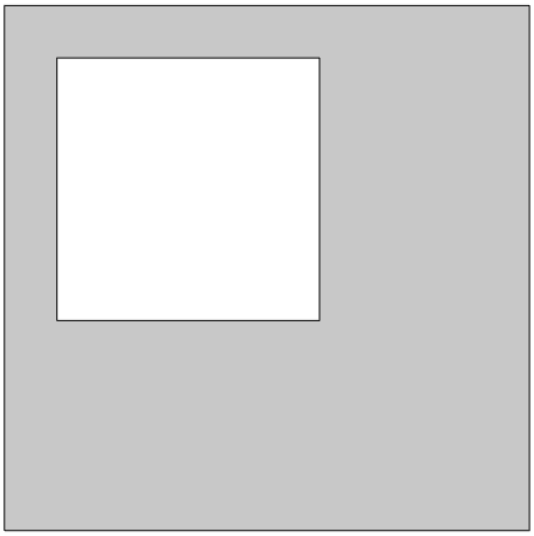


Figure 24: State 1 geometry in COMSOL.

A mesh is generated, and the study is solved.

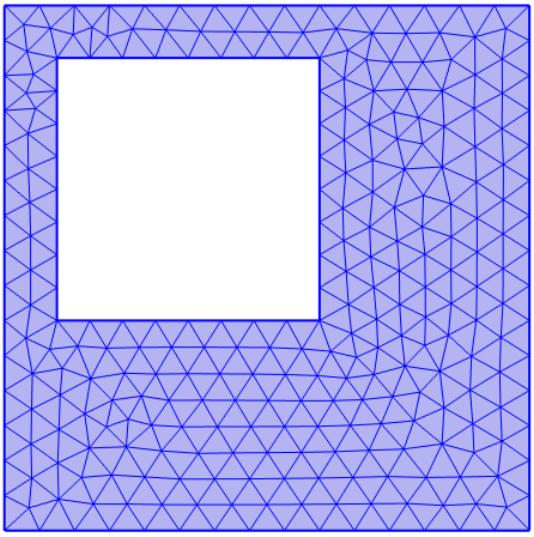


Figure 25: State 1 finite-element mesh.

The temperature distribution and isotherms are displayed.

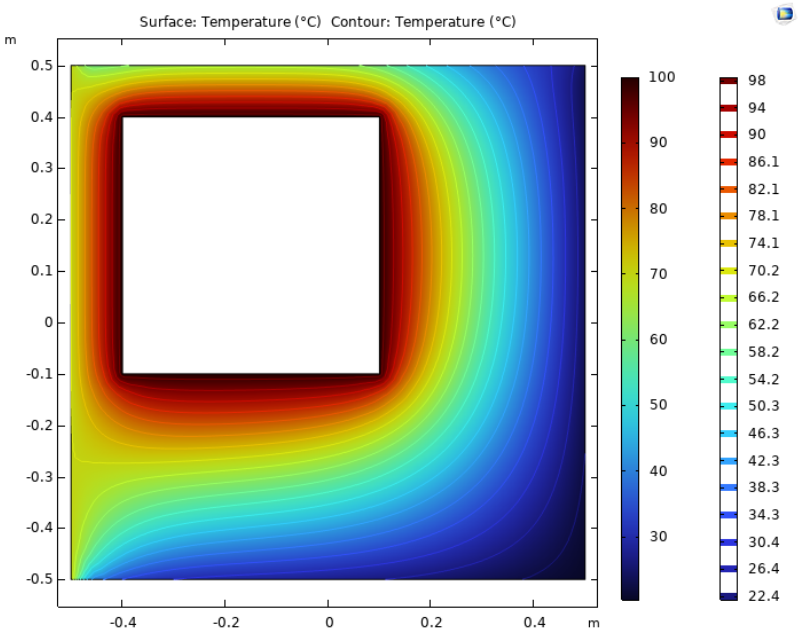


Figure 26: State 1 temperature contours and isotherms.

The source describes the COMSOL temperature range as approximately consistent with the MATLAB State 1 array. Several sample points are compared below. The original table's two group headings appear inconsistent with its data columns; the column labels below follow the row/column node versus coordinate distinction.

Table 1: Sample point comparison, source values.

MATLAB T	Column	Row	COMSOL T	Y	X
23.7631	11	1	23.8929	0.5	0.5
20.4224	11	11	20.4105	-0.5	0.5
22.111	11	9	22.044	-0.3	0.5
66.2026	3	9	65.3137	-0.3	-0.3

The source characterizes the sampled temperature differences as being around one percent or less and attributes differences to the coarser MATLAB grid. For heat transfer, it proposes a boundary line integration of normal total heat flux using the COMSOL expression ht.ntflux. The distinction between an integration and summing samples is reviewed in the appendix.

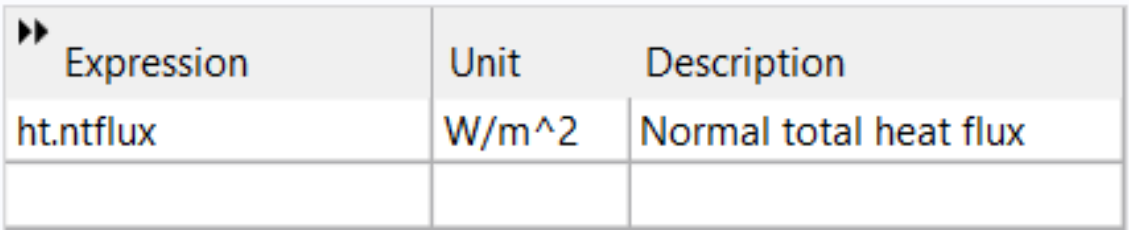


Figure 27: COMSOL normal total heat-flux expression.

The source states that a unit face area does not change its heat-rate value and reports 17292 W for COMSOL State 1 versus 18123 W for MATLAB, describing a discrepancy of about four percent. Its later comparison table reports a different COMSOL State 1 value; both are retained here for traceability.

The calculation is repeated for additional cavity locations.

For the source's State 3 illustration, the temperature and heat rate are obtained in the same way.

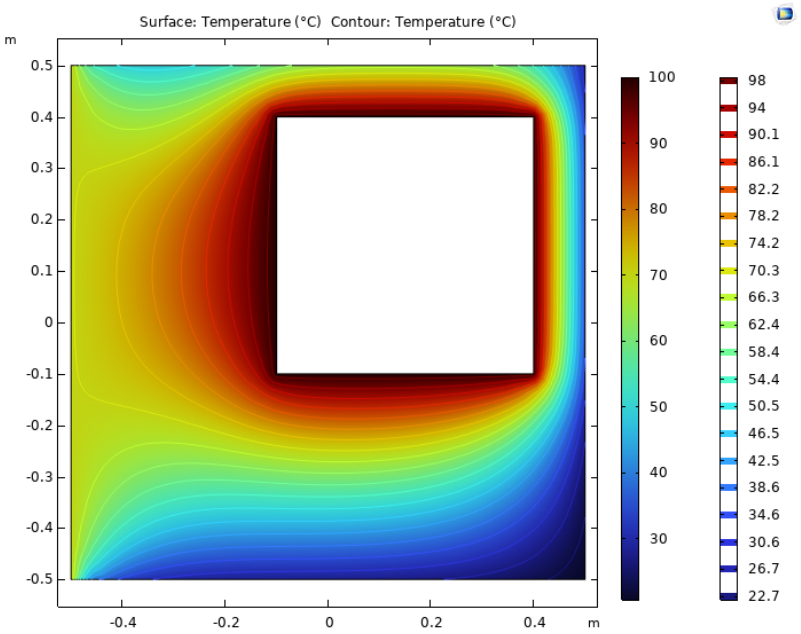


Figure 28: Temperature contour labeled State 3 in the source.

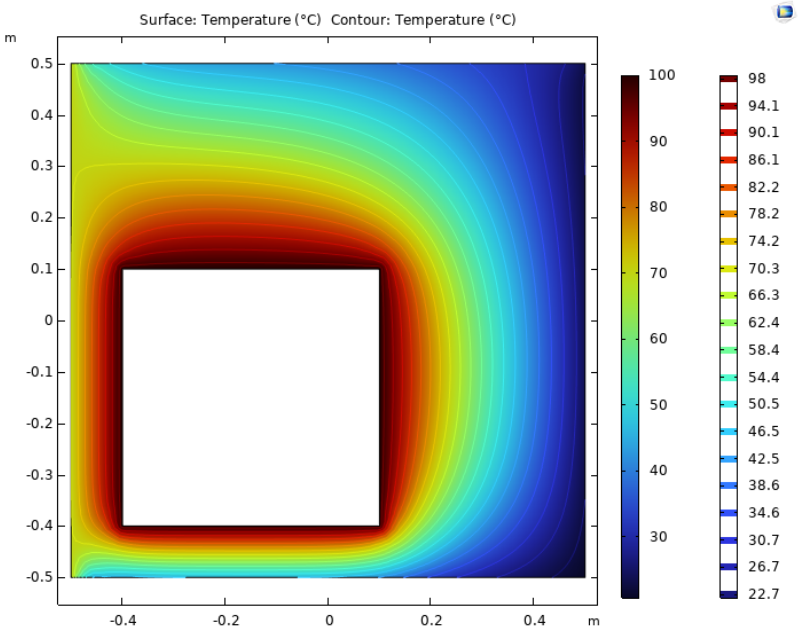


Figure 29: State 13 temperature contour.

The same procedure is shown for State 16.

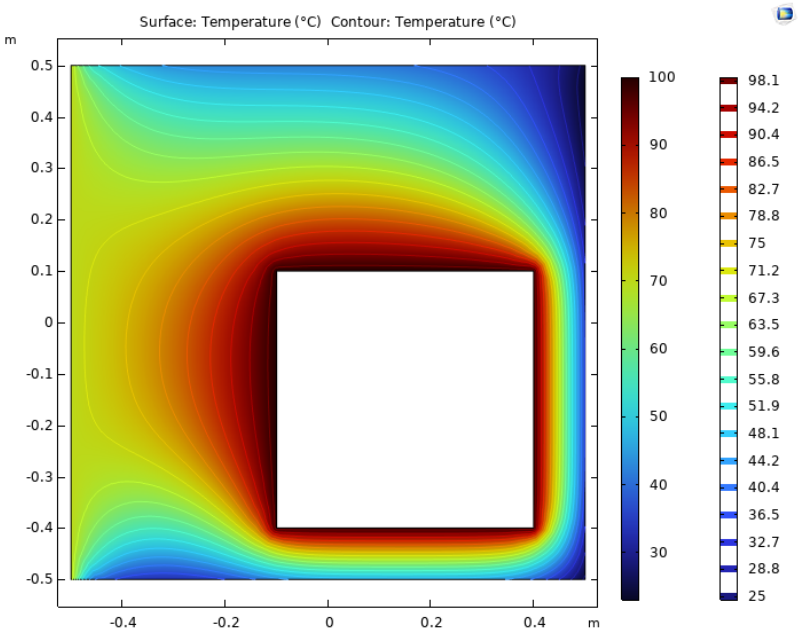


Figure 30: State 16 temperature contour.

The source observes uniform contour color along prescribed-temperature boundaries and isotherms locally parallel to them. The temperature distribution is then shown for State 5, the position selected by the MATLAB indicator.

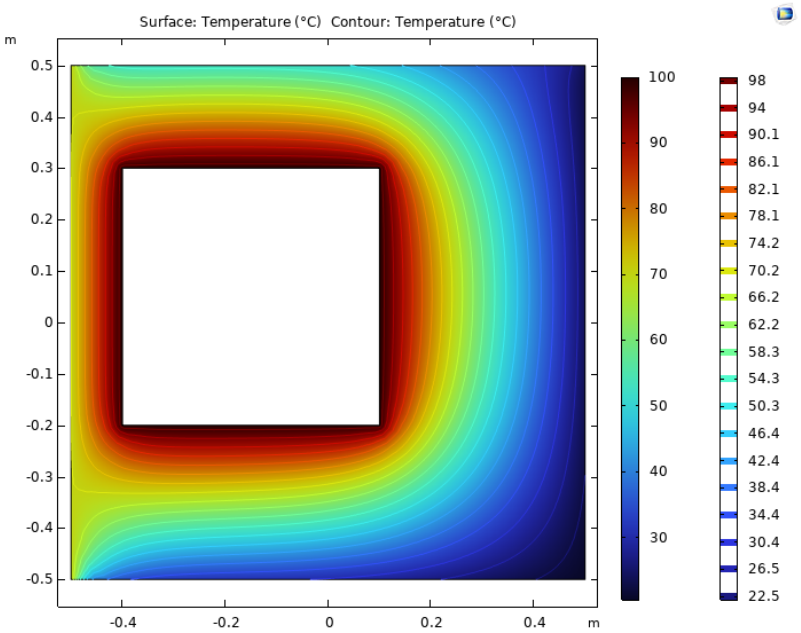


Figure 31: State 5 temperature contour.

State 8 is described as a cavity 20 cm below the top and 10 cm from the right wall.

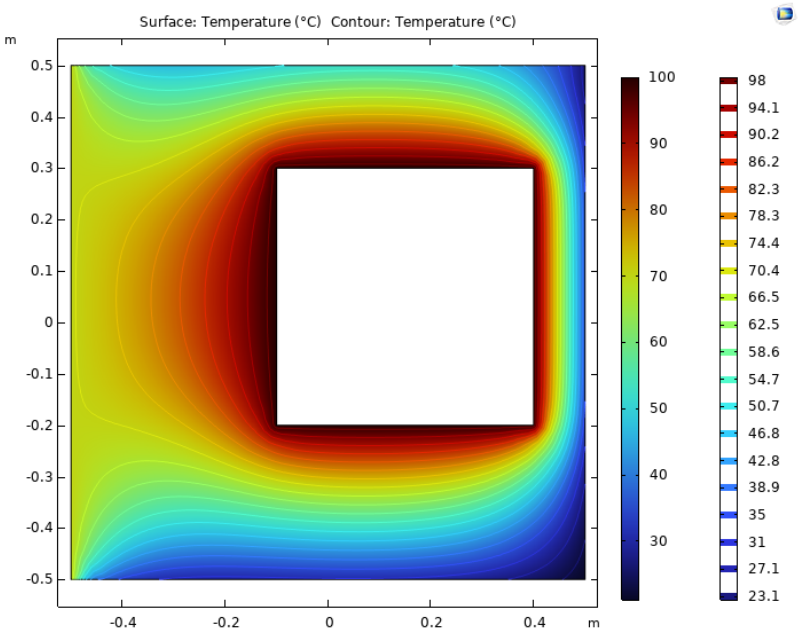


Figure 32: State 8 temperature contour.

State 9 is described as a cavity 20 cm above the bottom and 10 cm from the left wall.

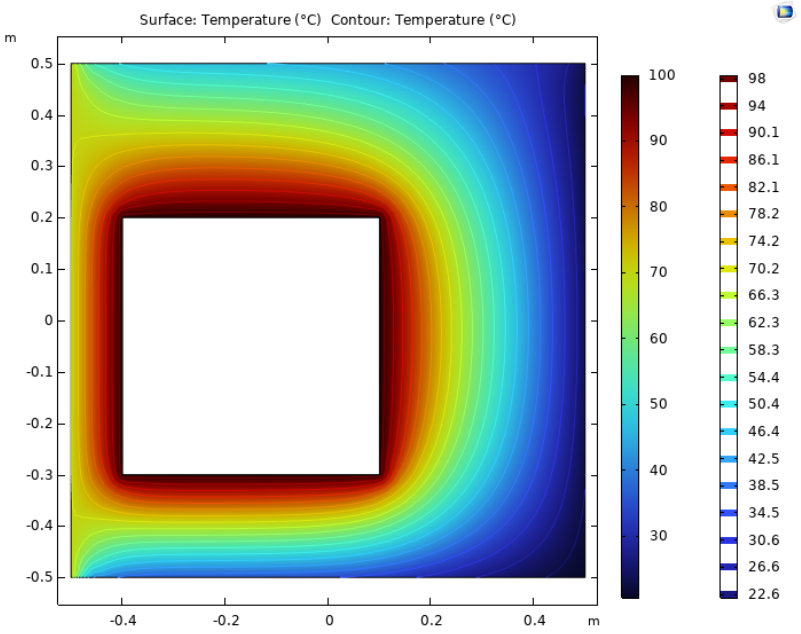


Figure 33: State 9 temperature contour.

State 12 is described as a cavity 20 cm above the bottom and 10 cm from the right wall.

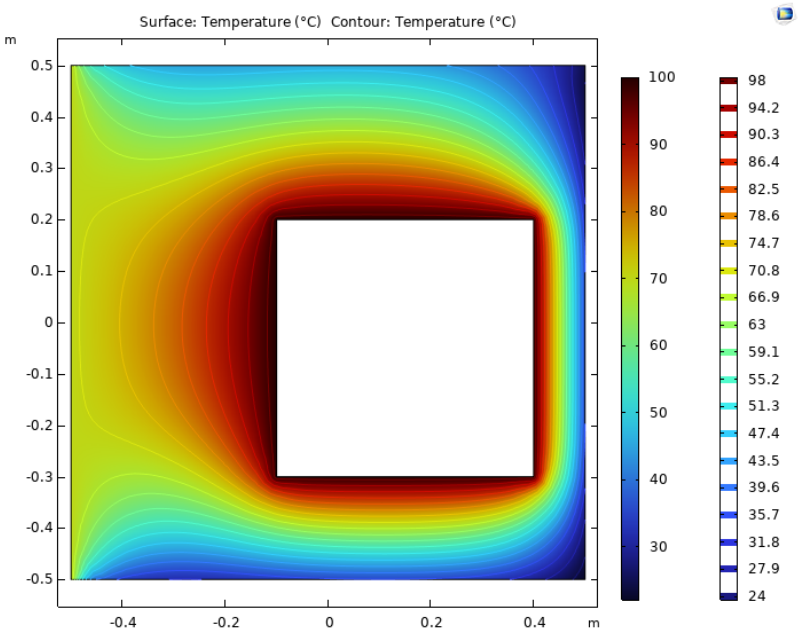


Figure 34: State 12 temperature contour.

The original reported heat-rate comparison is listed below.

Table 2: Historical heat-rate comparison, not corrected for the unit inconsistency.

COMSOL (kW)	MATLAB (kW)	State
-18.123	-18.1231	1
-17.76	-17.7513	4
-18.175	-18.1666	5
-17.845	-17.7889	8
-18.154	-18.1465	9
-17.773	-17.7664	12
-18.034	-18.0338	13
-17.648	-17.6528	16

The source selects State 5 as the best of the compared discrete configurations. These historical heat-rate results require correction and re-evaluation before that selection is used as an engineering design conclusion.

A State One COMSOL Point Data

The following table retains the source appendix's temperature samples for State 1. The coordinate origin is at the center of the chimney cross section. The original text and spreadsheet exports are also included in the repository.

X (m)	Y (m)	Temperature (°C)
-0.45326	-0.45691	54.30338
-0.5	-0.43333	70
-0.5	-0.5	70
-0.43333	-0.5	38.09135

X (m)	Y (m)	Temperature (°C)
-0.44805	-0.40007	61.42595
-0.5	-0.36667	70
-0.36667	-0.5	32.41545
-0.39682	-0.44038	48.42507
-0.44942	-0.33508	66.55718
-0.5	-0.3	70
-0.39856	-0.37138	59.40153
-0.33066	-0.44959	41.66711
-0.3	-0.5	30.3548
-0.35313	-0.39907	52.14934
-0.44821	-0.26869	70.39778
-0.5	-0.23333	70
-0.39618	-0.30749	66.58001
-0.44684	-0.2012	74.70343
-0.5	-0.16667	70
-0.2665	-0.4496	39.42791
-0.23333	-0.5	29.38578
-0.33834	-0.34289	60.12952
-0.29581	-0.39413	50.3269
-0.34727	-0.2777	69.46756
-0.30605	-0.30281	65.31366
-0.39403	-0.24168	73.62712
-0.38987	-0.19097	80.11538
-0.45929	-0.13367	77.88543
-0.5	-0.1	70
-0.45034	-0.07054	83.73102
-0.5	-0.03333	70
-0.45008	-0.00385	84.84022
-0.414	-0.15482	83.02815
-0.4	-0.0375	100
-0.5	0.033333	70
-0.45031	0.061197	84.88921
-0.4	0.025	100
-0.19994	-0.45067	37.94308
-0.16667	-0.5	28.80724
-0.23305	-0.39785	47.93508
-0.26811	-0.34058	58.51765
-0.2643	-0.28791	67.07892
-0.29464	-0.25176	73.14723
-0.3358	-0.21059	79.29199
-0.36061	-0.14864	89.02244
-0.4	-0.1	100
-0.5	0.1	70
-0.45179	0.125203	84.46072
-0.4	0.0875	100
-0.13343	-0.45094	36.97098
-0.1	-0.5	28.33797
-0.16589	-0.40194	45.92005
-0.20096	-0.34832	55.83952
-0.22004	-0.29932	64.54915
-0.23899	-0.24368	74.23179
-0.27158	-0.20027	81.72972
-0.30533	-0.15264	89.88742
-0.3375	-0.1	100
-0.06674	-0.45093	36.07558
-0.03333	-0.5	27.84294
-2.33E-04	-0.4507	35.06718

X (m)	Y (m)	Temperature (°C)
0.033333	-0.5	27.24973
-0.5	0.166667	70
-0.45453	0.18909	83.63367
-0.4	0.15	100
-0.09968	-0.40192	44.75782
-0.13068	-0.3531	53.68614
-0.16208	-0.30251	63.05948
-0.18711	-0.25211	72.26692
-0.21145	-0.20211	81.33266
-0.24322	-0.15121	90.55183
-0.275	-0.1	100
-0.0331	-0.40167	43.45626
0.032917	-0.40094	41.88311
0.066094	-0.45027	33.83816
0.1	-0.5	26.52489
-0.5	0.233333	70
-0.45541	0.256021	83.32658
-0.4	0.2125	100
-0.06457	-0.35218	52.40397
-0.09449	-0.30232	61.74365
-0.12298	-0.25186	71.34861
-0.15168	-0.2013	80.99589
-0.18159	-0.15082	90.53026
-0.2125	-0.1	100
6.71E-04	-0.3517	50.62794
0.065401	-0.35107	48.27281
0.098387	-0.40014	39.87681
0.132197	-0.44973	32.33581
0.166667	-0.5	25.66788
-0.5	0.3	70
-0.4409	0.303871	87.5043
-0.4	0.275	100
-0.45853	0.333287	81.89348
-0.03046	-0.30214	60.01595
-0.06039	-0.25212	69.89882
-0.08989	-0.20164	80.05144
-0.11968	-0.15091	90.15729
-0.15	-0.1	100
0.033333	-0.30202	57.54309
0.096682	-0.3016	54.18735
0.129842	-0.34988	45.31779
0.163682	-0.39888	37.47207
0.198415	-0.44903	30.57823
0.233333	-0.5	24.70351
-0.5	0.366667	70
-0.4	0.3375	100
-0.44469	0.368971	84.73389
0.002621	-0.25245	67.66213
-0.02734	-0.20204	78.47058
-0.05732	-0.1511	89.45052
-0.0875	-0.1	100
0.064943	-0.25309	64.2135
0.126323	-0.25277	59.57194
0.158764	-0.29973	50.18287
0.266807	-0.44851	28.52671
0.3	-0.5	23.66781
0.194256	-0.34724	41.91427

X (m)	Y (m)	Temperature (°C)
0.229496	-0.39715	34.70206
0.335751	-0.45222	26.0605
0.366667	-0.5	22.59398
-0.45698	0.412359	77.51995
-0.5	0.433333	70
-0.46342	0.463647	69.18242
-0.5	0.5	70
-0.4	0.4	100
0.035165	-0.20298	75.61103
0.005358	-0.15158	88.04667
-0.025	-0.1	100
0.095619	-0.20506	70.63709
0.151403	-0.20979	63.74577
0.181723	-0.25085	54.61513
0.217441	-0.2966	45.89947
0.303091	-0.39898	30.99512
0.26019	-0.34498	37.8933
0.368052	-0.3981	27.83669
0.39829	-0.44934	24.18694
0.433333	-0.5	21.50447
0.456569	-0.45544	22.12504
0.5	-0.5	20.41047
-0.41266	0.45677	74.28596
-0.43333	0.5	60.59833
-0.36908	0.444671	81.92363
-0.3375	0.4	100
0.066778	-0.15396	84.39056
0.0375	-0.1	100
0.12111	-0.15882	76.79338
0.170085	-0.17002	67.28704
0.203923	-0.20292	58.12776
0.239062	-0.24481	49.33625
0.28002	-0.29106	41.07527
0.33334	-0.33503	33.59523
0.39434	-0.35151	28.53306
0.439389	-0.39662	24.31465
0.5	-0.43333	20.958
-0.36667	0.5	62.98911
-0.33321	0.458587	78.28239
-0.30396	0.440974	85.11392
-0.275	0.4	100
0.1	-0.1	100
0.160956	-0.12447	75.63555
0.211211	-0.15147	63.0769
0.262801	-0.18673	52.01083
0.291415	-0.23644	44.4078
0.337329	-0.2662	37.60572
0.391867	-0.29664	31.09191
0.4475	-0.32986	25.59981
0.5	-0.36667	21.5046
-0.2125	0.4	100
-0.25617	0.455559	80.3042
-0.3	0.5	64.41057
0.155241	-0.06943	83.3806
0.1	-0.0375	100
0.208856	-0.09745	68.95458
0.25613	-0.12875	57.87397

X (m)	Y (m)	Temperature (°C)
0.300476	-0.14742	49.73185
0.329145	-0.19654	42.79613
0.386954	-0.23255	34.33552
0.444823	-0.26513	27.45725
0.5	-0.3	22.04405
-0.15	0.4	100
-0.19232	0.453543	81.17681
-0.23333	0.5	64.83684
0.15083	-0.00635	87.52672
0.1	0.025	100
0.203501	-0.0369	74.2827
0.252742	-0.06937	62.60305
0.301886	-0.09311	52.73471
0.345353	-0.12794	44.33209
0.389363	-0.16857	36.65783
0.44337	-0.19806	29.25168
0.5	-0.23333	22.56279
-0.0875	0.4	100
-0.12632	0.450596	82.20856
-0.16667	0.5	64.9288
0.146075	0.052109	89.59665
0.1	0.0875	100
0.197755	0.021231	77.6953
0.251992	-0.00526	65.73829
0.301507	-0.03612	55.3923
0.351038	-0.06769	45.93726
0.398643	-0.09923	37.71057
0.445485	-0.13233	30.41994
0.5	-0.16667	23.03935
-0.06145	0.450034	82.23831
-0.025	0.4	100
-0.1	0.5	64.86984
0.144407	0.113655	90.29912
0.1	0.15	100
0.19543	0.077719	79.1651
0.24807	0.051126	67.98889
0.300686	0.026093	57.40212
0.350479	-0.00377	47.89389
0.399705	-0.03535	39.12563
0.449312	-0.06743	31.00046
0.5	-0.1	23.4481
0.003439	0.449606	81.71983
0.0375	0.4	100
-0.03333	0.5	64.53596
0.149949	0.181451	88.98271
0.1	0.2125	100
0.146925	0.24367	89.0492
0.1	0.275	100
0.201369	0.144711	78.17736
0.242502	0.115587	69.76943
0.295656	0.078188	59.19026
0.351036	0.059394	48.89833
0.39998	0.030328	40.16084
0.449777	-0.00125	31.73729
0.5	-0.03333	23.76611
0.06866	0.448366	79.40389
0.1	0.4	100

X (m)	Y (m)	Temperature (°C)
0.033333	0.5	63.27869
0.201157	0.205266	77.7139
0.14657	0.360377	84.4937
0.1	0.3375	100
0.149262	0.305651	87.00013
0.194912	0.27343	77.41625
0.261882	0.188303	65.59357
0.29795	0.137998	59.02279
0.350087	0.122346	49.46972
0.400035	0.094907	40.67906
0.449668	0.064963	32.23451
0.5	0.033333	23.97876
0.153904	0.414787	74.16288
0.189246	0.39648	69.12939
0.132002	0.455912	68.48366
0.1	0.5	59.51144
0.247813	0.246075	67.20583
0.203397	0.337277	72.13164
0.241427	0.296993	66.55806
0.29599	0.23638	58.17651
0.333448	0.192712	52.09383
0.398619	0.163928	40.90923
0.450043	0.129907	32.32158
0.5	0.1	24.08083
0.236373	0.39216	60.34488
0.200416	0.44758	58.85298
0.166667	0.5	52.75198
0.288795	0.286515	57.93597
0.266441	0.340564	59.24434
0.337451	0.269532	49.87056
0.396416	0.230946	40.67979
0.450249	0.196883	32.10738
0.5	0.166667	24.0715
0.296201	0.392049	50.57699
0.264617	0.446571	49.82981
0.233333	0.5	45.98845
0.333745	0.337493	48.01086
0.394479	0.304045	39.61761
0.447434	0.265465	32.01017
0.5	0.233333	23.94989
0.351061	0.396986	42.5225
0.329097	0.448085	42.03526
0.3	0.5	40.27347
0.396411	0.368087	37.54402
0.448244	0.332787	31.00315
0.5	0.3	23.72017
0.395905	0.439885	35.19481
0.366667	0.5	35.29074
0.447813	0.398161	30.01233
0.5	0.366667	23.41358
0.454748	0.456318	28.50307
0.433333	0.5	30.38475
0.5	0.433333	23.17201
0.5	0.5	23.89287

B Editorial Review

The source uses TC/Vis in Celsius for most convective surface terms but retains ambient T1, T2, and T3 in kelvin. Temperature differences must use a consistent scale. Some corner terms use T4 in kelvin while others use Vis in Celsius. The resulting sum therefore cannot be treated as a physical heat rate. This issue affects both supplied MATLAB scripts and the reported State 5 selection. Their numerical logic is preserved rather than silently repaired.

The outer heat-transfer sum also includes convection on the top, bottom, and right walls, but does not include conductive exchange through the prescribed-temperature left wall. Thus, even after correcting units, it should not automatically be called total heat transfer through the entire outer boundary. An independent energy balance and a declared objective are needed.

The code variable maxq is assigned min(Qall), selecting the most negative signed value. Largest signed value, largest outward heat rate, and largest magnitude are different objectives. The sign convention must be established before choosing an optimum. A coarse sixteen-point placement search is not proof of a continuous global optimum; no mesh-convergence study is supplied.

The report's COMSOL State 1 prose quotes 17292 W, while its final table quotes -18.123 kW. The prose also discusses State 3, whereas the final table lists State 4. These inconsistencies must be reconciled with the saved models and exact boundary selections. Small differences in selected temperatures do not validate an inconsistent heat-rate calculation.

COMSOL ht.ntflux is a normal heat-flux quantity. To obtain heat rate for a two-dimensional section, integrate along the specified boundary with the intended out-of-plane depth and sign convention. Summing point samples is not generally the same as boundary integration. Inspect the saved evaluation method before accepting the historical comparison values.

The code fills the whole cavity block with the prescribed gas temperature, including interior placeholder nodes, but those nodes are not physical brick temperatures. Distinguish placeholders from material-domain results. Density and heat capacity are listed material properties, but they do not enter a steady constant-conductivity temperature equation without transient storage.

The native archive includes eleven COMSOL MPH models, two MATLAB Live Scripts, and text/spreadsheet point exports. It does not include a native model for every one of the sixteen placements. The exported point table and the source appendix may use different precision or sampling; both are preserved independently. The two extracted M files are text versions of the supplied code, not corrected implementations. Neither MATLAB nor COMSOL was executed during conversion. No solver validation, physical experiment, or manufacturing design approval is claimed.